

Supplementary File 5. Data Charting Form

Review title

Reframing the malaria-sickle cell disease relationship in sub-Saharan Africa as a bounded syndemic: A scoping review and evidence gap map

Purpose

This data charting form specifies the variables to be extracted from each included source. It is designed to capture descriptive study characteristics, biological evidence, spatial evidence, clinical evidence, health-system visibility, structural vulnerability, programme response, and syndemic theory use.

The form should be piloted on a sample of included sources before full extraction. Fields may be refined after pilot charting if the review team identifies ambiguity or missing variables.

Data charting instructions

Each included source should be assigned a unique study ID. Reviewers should extract information directly from the full text wherever possible. If a field is not reported, enter **Not reported**. If a field is unclear, enter **Unclear** and add a reviewer note.

Where one study contributes to more than one evidence domain, reviewers should code all applicable domains. For example, a study may contribute to biological interaction, clinical convergence, and service-mediated visibility.

Table 1. Core data charting form

Field number	Variable	Description or coding instruction	Response format
1	Study ID	Unique identifier assigned by review team.	Free text

2	Reviewer name	Name or initials of reviewer extracting data.	Free text
3	Extraction date	Date data were charted.	Date
4	Author(s)	First author and co-authors as reported.	Free text
5	Year	Year of publication or report.	Year
6	Title	Full title of article or report.	Free text
7	Source type	Type of evidence source.	Journal article, review, report, policy document, thesis, conference abstract, programme document, other
8	Journal or organisation	Journal name or publishing organisation.	Free text
9	DOI or URL	Persistent identifier where available.	Free text
10	Peer-reviewed	Whether the source was peer-reviewed.	Yes, No, Unclear
11	Language	Language of publication.	Free text
12	Country or countries	SSA country or countries included.	Free text

13	SSA sub-region	Geographic sub-region.	West Africa, East Africa, Central Africa, Southern Africa, Multiregional SSA, Africa-wide, Global with SSA data
14	Study location	Specific setting, city, district, region, or facility.	Free text
15	Urban or rural setting	Setting classification.	Urban, Rural, Mixed, Not reported
16	Spatial scale	Scale of evidence.	Facility, Community, District, Region, National, Multicountry, Continental, Global
17	Study aim	Stated aim or objective of the source.	Free text
18	Study design	Study design or evidence type.	Cohort, Case-control, Cross-sectional, Clinical trial, Modelling, Geospatial, Routine data analysis, Programme evaluation,

			Implementation study, Review, Policy analysis, Other
19	Study period	Dates or years of data collection.	Free text
20	Population group	Population studied.	Neonates, Infants, Children, Adolescents, Adults, Pregnant women, Caregivers, Health workers, Facilities, Programmes, Mixed
21	Age range	Age range or age group as reported.	Free text
22	Sample size	Number of participants, records, facilities, locations, or reports.	Numeric or free text
23	Comparator group	Comparator or reference group, if applicable.	Free text
24	Data source	Source of data.	Survey, Hospital records, Laboratory data, Routine health information system, Registry, Screening

			database, Trial data, Programme records, Modelled data, Literature review, Other
25	Funding source	Funding source for the included study, if reported.	Free text
26	Conflicts of interest	Conflicts reported by study authors.	Free text or Not reported

Table 2. Biological and genetic variables

Field number	Variable	Description or coding instruction	Response format
27	SCD included	Whether study included SCD.	Yes, No, Unclear
28	SCA or HbSS included	Whether study included sickle cell anaemia or HbSS.	Yes, No, Unclear
29	HbAS or sickle cell trait included	Whether study included HbAS or sickle cell trait.	Yes, No, Unclear
30	HbS frequency reported	Whether HbS frequency or prevalence was reported.	Yes, No
31	SCD prevalence reported	Whether SCD prevalence was reported.	Yes, No

32	SCA birth burden reported	Whether estimated or observed SCA births were reported.	Yes, No
33	Other red blood cell polymorphisms	Other polymorphisms studied.	HbC, Alpha-thalassaemia, G6PD, ABO, Dantu, Other, None
34	Genotype ascertainment method	Method used to identify genotype or phenotype.	HPLC, IEF, Electrophoresis, PCR, Genotyping array, Sequencing, Point-of-care test, Clinical diagnosis, Not reported
35	Confirmatory testing used	Whether confirmatory testing was reported.	Yes, No, Unclear
36	Biological mechanism assessed	Mechanism studied.	HbAS protection, Parasite invasion, Rosetting, Cytoadherence, Parasite growth, Erythrocyte membrane tension, Immune response, Epistasis, Not applicable

37	Malaria protection outcome	Malaria outcome linked to genotype or polymorphism.	Parasitaemia, Uncomplicated malaria, Severe malaria, Severe malarial anaemia, Cerebral malaria, Respiratory distress, Mortality, Other
38	Main biological finding	Summary of biological or genetic result.	Free text
39	Evidence of epistasis	Whether co-inheritance or interaction between polymorphisms was assessed.	Yes, No, Unclear
40	Epistasis finding	Finding related to co-inheritance, if applicable.	Free text

Table 3. Malaria variables

Field number	Variable	Description or coding instruction	Response format
41	Malaria included	Whether malaria was included.	Yes, No
42	Malaria species	Species assessed.	P. falciparum, Mixed species, Not reported

43	Malaria diagnostic method	Method used to diagnose malaria.	Microscopy, RDT, PCR, Clinical diagnosis, Routine data, Modelled exposure, Not reported
44	Malaria exposure measure	How exposure was measured.	Parasite prevalence, Incidence, Mortality, Endemicity, Transmission intensity, Ecological suitability, Modelled risk, Facility-reported cases, Other
45	Parasite density reported	Whether parasite density was reported.	Yes, No
46	PfHRP2 or parasite biomass reported	Whether PfHRP2 or similar marker was reported.	Yes, No
47	Severe malaria definition	Definition used by study.	Free text
48	Malaria phenotype	Phenotype assessed.	Uncomplicated malaria, Severe malaria, Severe malarial anaemia, Cerebral malaria,

			Respiratory distress, Malaria-positive illness, Fever with parasitaemia, Other
49	Malaria outcome	Main malaria-related outcome.	Free text
50	Malaria-SCD comparison	Whether malaria outcomes were compared by SCD or genotype status.	Yes, No
51	Main malaria finding	Summary of malaria-related result.	Free text

Table 4. Clinical convergence variables

Field number	Variable	Description or coding instruction	Response format
52	Severe anaemia assessed	Whether severe anaemia was assessed.	Yes, No
53	Severe anaemia threshold	Haemoglobin threshold used.	Free text
54	Baseline haemoglobin reported	Whether baseline Hb was reported for SCD or comparison group.	Yes, No

55	Acute febrile illness assessed	Whether fever or febrile illness was assessed.	Yes, No
56	Respiratory distress assessed	Whether respiratory distress was assessed.	Yes, No
57	Acute chest syndrome assessed	Whether acute chest syndrome was assessed.	Yes, No
58	Transfusion need assessed	Whether transfusion need or transfusion received was assessed.	Yes, No
59	Hospital admission assessed	Whether hospital admission data were used.	Yes, No
60	Mortality assessed	Whether mortality was assessed.	Yes, No
61	Diagnostic attribution assessed	Whether attribution between malaria, SCD, anaemia, or other syndromes was assessed.	Yes, No, Unclear
62	SCD under-recognition assessed	Whether missed or previously unknown SCD was assessed.	Yes, No

63	Previously unknown SCD reported	Whether previously unknown SCD or SCA was identified.	Yes, No, Not applicable
64	Malaria-positive SCD illness reported	Whether malaria-positive illness among SCD or SCA patients was reported.	Yes, No
65	Main clinical convergence finding	Summary of clinical finding.	Free text

Table 5. Spatial and geographic variables

Field number	Variable	Description or coding instruction	Response format
66	Spatial analysis included	Whether spatial analysis was included.	Yes, No
67	Spatial method	Type of spatial method used.	Descriptive mapping, Geostatistical modelling, Cluster analysis, Spatial autocorrelation, Catchment analysis, Accessibility analysis, Facility mapping, Other
68	Geographic unit	Spatial unit of analysis.	Facility, Village, Parish, District,

			Region, Country, Grid cell, Catchment, Other
69	Malaria map or spatial measure	Whether malaria was mapped or spatially modelled.	Yes, No
70	HbS or SCD map or spatial measure	Whether HbS, SCD, or SCA births were mapped or spatially modelled.	Yes, No
71	Spatial overlap assessed	Whether overlap between malaria and HbS or SCD was assessed.	Yes, No
72	Formal co- clustering assessed	Whether formal spatial co-clustering was assessed.	Yes, No
73	Service access mapped	Whether access to SCD services was mapped.	Yes, No
74	Facility readiness mapped	Whether facility readiness was mapped.	Yes, No
75	Main spatial finding	Summary of spatial result.	Free text
76	Planning scale relevance	Whether results are relevant to planning scale.	National, Subnational, District, Facility catchment,

			Programme network, Not applicable
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Table 6. Service-mediated visibility variables

Field number	Variable	Description or coding instruction	Response format
77	Newborn screening included	Whether newborn screening was studied.	Yes, No
78	Screening platform	Screening platform or method.	Dried blood spot, HPLC, IEF, Point-of-care test, EID platform, Other, Not reported
79	Confirmatory diagnosis included	Whether confirmatory diagnosis was included.	Yes, No
80	Result return assessed	Whether return of results was assessed.	Yes, No
81	Caregiver notification assessed	Whether caregiver notification was assessed.	Yes, No
82	Counselling assessed	Whether counselling was assessed.	Yes, No

83	Clinic enrolment assessed	Whether enrolment into care was assessed.	Yes, No
84	Routine data source used	Whether routine data were used.	Yes, No
85	Routine data system	Type of routine data system.	HMIS, DHIS2, Hospital records, Clinic register, Laboratory register, Mortality data, Transfusion records, Other
86	Coding or reporting assessed	Whether diagnostic coding or reporting was assessed.	Yes, No
87	Mortality attribution assessed	Whether cause of death or attribution was assessed.	Yes, No
88	Service-visible burden discussed	Whether study distinguishes recorded burden from underlying burden.	Yes, No
89	Hidden burden discussed	Whether hidden or undercounted SCD burden was discussed.	Yes, No

90	Main visibility finding	Summary of visibility-related result.	Free text
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Table 7. Structural vulnerability and essential care variables

Field number	Variable	Description or coding instruction	Response format
91	Structural vulnerability addressed	Whether structural vulnerability was addressed.	Yes, No
92	Access to diagnosis assessed	Whether access to diagnosis was assessed.	Yes, No
93	Penicillin prophylaxis assessed	Whether penicillin prophylaxis was assessed.	Yes, No
94	Vaccination assessed	Whether vaccination was assessed.	Yes, No
95	Hydroxyurea assessed	Whether hydroxyurea access, use, safety, or effect was assessed.	Yes, No
96	Transfusion readiness assessed	Whether transfusion readiness or availability was assessed.	Yes, No

97	Specialist care assessed	Whether access to specialist SCD care was assessed.	Yes, No
98	Referral assessed	Whether referral systems were assessed.	Yes, No
99	Follow-up or retention assessed	Whether follow-up or retention was assessed.	Yes, No
100	Geographic access assessed	Whether distance, travel time, transport, or accessibility was assessed.	Yes, No
101	Workforce or training assessed	Whether workforce or training was assessed.	Yes, No
102	Medicine supply assessed	Whether medicine supply was assessed.	Yes, No
103	Laboratory capacity assessed	Whether laboratory capacity was assessed.	Yes, No
104	Main structural vulnerability finding	Summary of structural vulnerability result.	Free text

Table 8. Programme response variables

Field number	Variable	Description or coding instruction	Response format
105	Programme response included	Whether programme response was studied.	Yes, No
106	Programme type	Type of programme.	Newborn screening, Point-of-care testing, Hydroxyurea programme, Comprehensive care, Transfusion programme, Referral network, Implementation network, Other
107	Programme country or countries	Country or countries involved.	Free text
108	Screening-to-care cascade described	Whether screening-to-care cascade was described.	Yes, No
109	Cascade stages reported	Stages reported.	Screen, Confirm, Notify, Counsel, Enrol, Prophylaxis, Hydroxyurea assessment, Transfusion access,

			Refer, Retain, Mortality ascertainment
110	Point-of-care testing assessed	Whether point-of-care testing was assessed.	Yes, No
111	Decentralised care assessed	Whether decentralised care was assessed.	Yes, No
112	Hub-and-spoke system assessed	Whether hub-and-spoke or networked care was assessed.	Yes, No
113	Implementation barriers	Barriers reported.	Free text
114	Implementation facilitators	Facilitators reported.	Free text
115	Sustainability addressed	Whether sustainability was addressed.	Yes, No
116	Main programme response finding	Summary of programme result.	Free text

Table 9. Syndemic evidence coding

Field number	Variable	Description or coding instruction	Response format
117	Explicit syndemic theory used	Whether the source explicitly uses syndemic theory or terminology.	Yes, No

118	Implicit syndemic logic present	Whether the source links evidence in a syndemic-relevant way without using the term.	Yes, No, Unclear
119	Co-occurrence code	Malaria and SCD or HbS occur in same population or setting.	Yes, No
120	Spatial co-risk code	Malaria and HbS, SCD, or SCA burden are spatially patterned or overlapping.	Yes, No
121	Biological interaction code	Biological interaction between malaria and red blood cell traits.	Yes, No
122	Clinical convergence code	Clinical overlap through severe anaemia, fever, malaria-positive illness, transfusion, or attribution.	Yes, No
123	Service-mediated visibility code	Recorded burden shaped by screening, diagnosis, coding, reporting, follow-	Yes, No

		up, or mortality attribution.	
124	Structural vulnerability code	Unequal access to care modifies outcomes, recognition, or survival.	Yes, No
125	Programme response code	Programme response through screening, testing, care linkage, referral, or implementation.	Yes, No
126	Level of evidence integration	Degree to which study integrates domains.	Level 1, Level 2, Level 3, Level 4
127	Type of syndemic relationship supported	Main relationship supported by the study.	Co-occurrence, Spatial, Biological, Clinical, Visibility, Structural, Programme, Integrated
128	Does study support bounded syndemic model?	Reviewer judgement based on evidence.	Yes, Partially, No, Unclear
129	Reviewer justification for syndemic coding	Brief explanation for coding decision.	Free text

Table 10. Findings, gaps, and reviewer notes

Field number	Variable	Description or coding instruction	Response format
130	Main finding relevant to review	Concise statement of key finding.	Free text
131	Domain-specific finding	Finding linked to assigned domain.	Free text
132	Author-reported limitations	Limitations reported by study authors.	Free text
133	Reviewer-identified limitations	Limitations noted by reviewer.	Free text
134	Evidence gap identified by source	Gap stated or implied by included source.	Free text
135	Relevance to review question	How the source contributes to review question.	Free text
136	Relevance to programme planning	Whether and how source informs screening or care planning.	Free text
137	Include in domain evidence table	Whether source should appear in Supplementary File 9.	Yes, No
138	Notes for synthesis	Reviewer notes for narrative or	Free text

		configurative synthesis.	
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Coding guidance for level of evidence integration

Level	Definition	Coding example
Level 1	Single-domain evidence.	Study examines HbAS protection against malaria only.
Level 2	Links two domains.	Study links severe anaemia admissions with SCA genotype.
Level 3	Links three or more domains.	Study links SCA, severe anaemia, malaria positivity, and clinical under-recognition.
Level 4	Syndemic-integrated evidence.	Study links biological risk, spatial co-risk, clinical convergence, service visibility, structural vulnerability, and programme response.

Data charting quality control

The review team should use the following quality checks:

1. Pilot the form on a sample of included studies.
2. Compare extraction across reviewers.
3. Resolve differences through discussion.

4. Refine unclear fields before full charting.
5. Maintain a version-controlled extraction file.
6. Document changes to the charting form.
7. Retain reviewer notes for ambiguous sources.

Version control log

Version	Date	Change made	Reason	Approved by
1.0	To be inserted	Initial data charting form	Initial version	To be inserted
1.1	To be inserted	To be inserted	To be inserted	To be inserted